

Learning Day 6 Autoencoders

Top-to-Bottom Workflow Flowchart with Short Notes

This document presents the complete workflow for the three completed official LD6 notebooks. It is intended as a quick visual guide to the full project path from theory to final interpretation.

Learning Day 6 Autoencoders: Top-to-Bottom Workflow

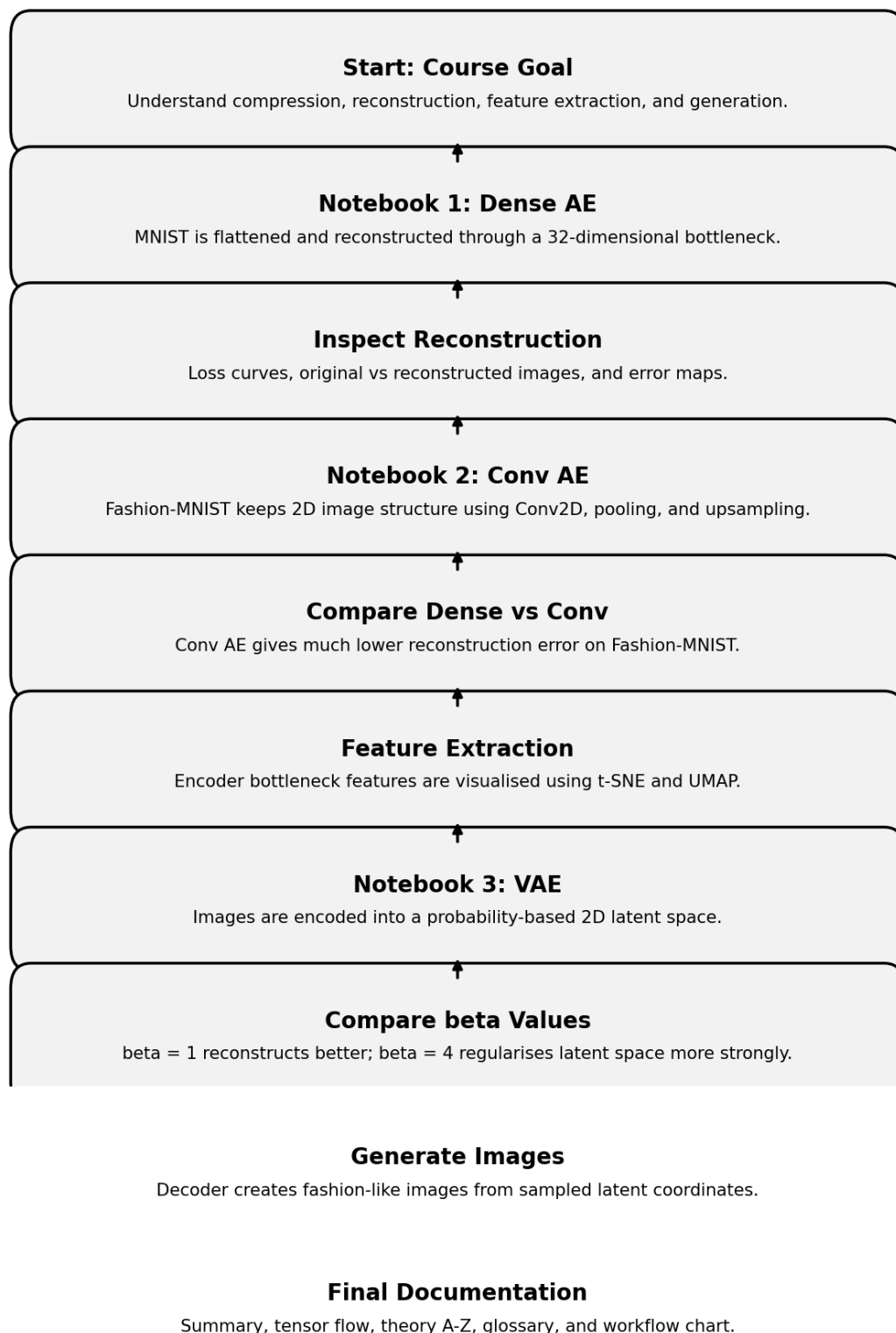


Figure 1: Complete top-to-bottom workflow for the completed LD6 Autoencoder notebooks.

Step Notes

Step	Short Note	Main Output
Course Goal	Understand reconstruction, compression, feature extraction, and generation.	Clear learning objective.
Dense Autoencoder	Flatten MNIST images and reconstruct them through a 32-dimensional bottleneck.	MNIST reconstructions and error maps.
Conv Autoencoder	Use Conv2D and pooling to preserve image structure on Fashion-MNIST.	Better Fashion-MNIST reconstructions.
Dense vs Conv	Compare reconstruction error on the same test images.	Conv AE reduced error by about 82.3%.
Feature Extraction	Use encoder bottleneck features and visualise with t-SNE/UMAP.	Class grouping insight.
VAE beta = 1	Train standard VAE balance.	Better VAE reconstruction.
VAE beta = 4	Train stronger latent-space regularisation.	Lower KL loss, blurrier output.
Latent Generation	Decode sampled points from $[-3, 3] \times [-3, 3]$.	Generated fashion-like images.
Final Documentation	Create summary, tensor flow, theory A-Z, glossary, and workflow documents.	Complete learning record.

Final Interpretation

The three notebooks show the progression from simple reconstruction to image-aware reconstruction and finally to generation. Dense Autoencoders are best for learning the core idea, Convolutional Autoencoders are better for images, and VAEs add probabilistic latent-space learning so new images can be generated.

Supplementary Work Later

The remaining practical notebooks for sparse autoencoders, denoising autoencoders, anomaly detection, and ECG/time-series autoencoders will be created later as separate focused projects.